Hard-Level Array DSA Questions (Java)

1. Maximum Product Subarray
Given an array of integers, find the contiguous subarray that has the largest product.
Example: Input: nums = [2,3,-2,4] → Output: 6

2. Median of Two Sorted Arrays
Find the median of two sorted arrays. The overall run time complexity should be O(log(min(m,n))).

3. Trapping Rain Water
Given n non-negative integers representing elevation map, compute how much water it can trap after raining.
Example: Input: height = [0,1,0,2,1,0,1,3,2,1,2,1] → Output: 6

4. Largest Rectangle in Histogram
Find the area of the largest rectangle in a histogram represented by an array of heights.

5. Sliding Window Maximum
Given an array and a sliding window size k, return the max element in each window.
Example: Input: nums = [1,3,-1,-3,5,3,6,7], k = 3 → Output: [3,3,5,5,6,7]

6. Subarray Sum Equals K
Count the number of continuous subarrays that sum up to k using prefix sum + hashmap.
Example: Input: nums = [1,1,1], k = 2 → Output: 2

7. Kth Largest Element in an Array
Find the kth largest element in an unsorted array in O(n) average time.
Example: Input: nums = [3,2,1,5,6,4], k = 2 → Output: 5

8. Count Inversions in an Array
An inversion is a pair (i, j) such that i < j and arr[i] > arr[j]. Count total inversions using merge sort.

9. Maximum Subarray Sum After One Deletion
You can delete one element. Find the maximum subarray sum possible after deletion.

10. Merge Intervals
Given a list of intervals, merge overlapping intervals.
Example: Input: [[1,3],[2,6],[8,10],[15,18]] → Output: [[1,6],[8,10],[15,18]]

11. Maximum Sum of Non-Adjacent Elements
Find the maximum sum of non-adjacent elements (like the House Robber problem).
Example: Input: [3,2,5,10,7] → Output: 15

12. Find the Longest Subarray with Equal Number of 0s and 1s
Convert 0s to -1 and use prefix sum technique.

13. Minimum Number of Swaps to Sort the Array
Using visited array and cycle detection, find the minimum swaps to sort the array.

14. Longest Consecutive Sequence
Find the length of the longest sequence of consecutive elements in O(n) time.
Example: Input: [100, 4, 200, 1, 3, 2] → Output: 4

15. Find All Duplicates in an Array in O(n) Time & Constant Space
Mark visited indices by negating values to find duplicates.